CRYPTONITE

RESEARCH TASK PHASE 1

Module 2: Decision Trees, Forests & Gradient Boosters

Objective & Core Topics

This task is designed to build a practical and theoretical foundation in tree-based architectures and gradient boosting algorithms:

- Decision Trees (Splitting metrics, recursive partitioning)
- Random Forests (Bagging, feature subsampling)
- Gradient Boosting: XGBoost, LightGBM, and CatBoost

Dataset

- **Cybersecurity Intrusion Detection Dataset:** Kaggle Dataset Link
- *Note:* As part of your EDA, explicitly verify class distribution and missing values.

Task List

1. **Exploratory Data Analysis (EDA):** Conduct an in-depth analysis to check feature relationships, distributions, mising values, and class imbalance to make your feature engineering pipeline.
2. **Baseline Model:** Train a baseline Random Forest on a train/validation split. Report Accuracy, F1 score, ROC-AUC, and Confusion Matrix.
   - *Reference Baseline Target:* Accuracy ≥ 0.8831, F1 ≥ 0.8508, ROC-AUC ≥ 0.8762.
3. **Feature Engineering:** Improve the baseline using at least **three** of the following methods, reporting score deltas for each:
   - Out-of-fold target encoding or frequency encoding for high-cardinality categories.
   - Interaction features (e.g., numeric ratios, differences).
   - Aggregation statistics (mean/median/count grouped by categories).
   - Look for outliers and missing values imputation.
4. **Report (1–2 Pages):** Write a structured summary covering:
   - Mathematical formulation of Decision Trees (Gini impurity, Entropy, Information Gain) and their intuitive mechanics.
   - Comparative performance analysis across feature engineering techniques.
   - (If attempting Bonus) Boosting implementation details.

Kaggle Playground Competition

- Join the latest Kaggle Playground competition, deploy your model, and optimize your leaderboard ranking.
- Share your Kaggle username to allow leaderboard tracking.
- Start working on kaggle only after completing the tasks.

Bonus Task (Optional)

- Train and tune **XGBoost**, **LightGBM**, and **CatBoost** models on the intrusion dataset.

Evaluation and Submission Guidelines

- Create a `Task-2` folder in the GitHub repository containing:
  - `Model_Tuning.ipynb`
  - `Kaggle_Competition_and_Boosting.ipynb` (if attempting bonus)
  - `Decision_Trees_and_Boosters_Report.pdf`
  - `README.md`
- **Evaluation Criteria:** Model performance, depth of conceptual and mathematical understanding of tree partitioning, feature engineering efficacy, and surpassing baseline metrics.
- **TASK-2 DEADLINE: 17th October, 2026**

Resources

- Random Forest From Scratch (deep-ml)
- Random Forest vs. Gradient Boosting
- Decision Trees (CampusX Playlist)
- CatBoost vs. XGBoost vs. LightGBM
- Kaggle Competitions: A Beginner's Guide

Points of Contact (Message Only)

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